

Industry & Operational Management

IMG 401

Department of Computer Science & Engineering
University of Asia Pacific



Welcome, Future Innovators!

I'm Muhammad Abdullah Al Muti (Farhan), your instructor for Industry and Operational Management.



B. Sc. in CSE, 2004



Consultant
Strategy & Business Development



Head of Customer Success Mgt.
Verbex AI



Strategic Partnerships Manager (Former)
Telco Ecosystem Partnerships, APAC, Singapore



Deputy General Manager (Former)
Head of Web, Payments & Global Partnerships



- 19+ years of telecom and tech industry expertise, specializing in digital transformation, product management & business development
- Spearheaded the successful launch & commercialization of 10+ high-impact digital platforms over the past decade

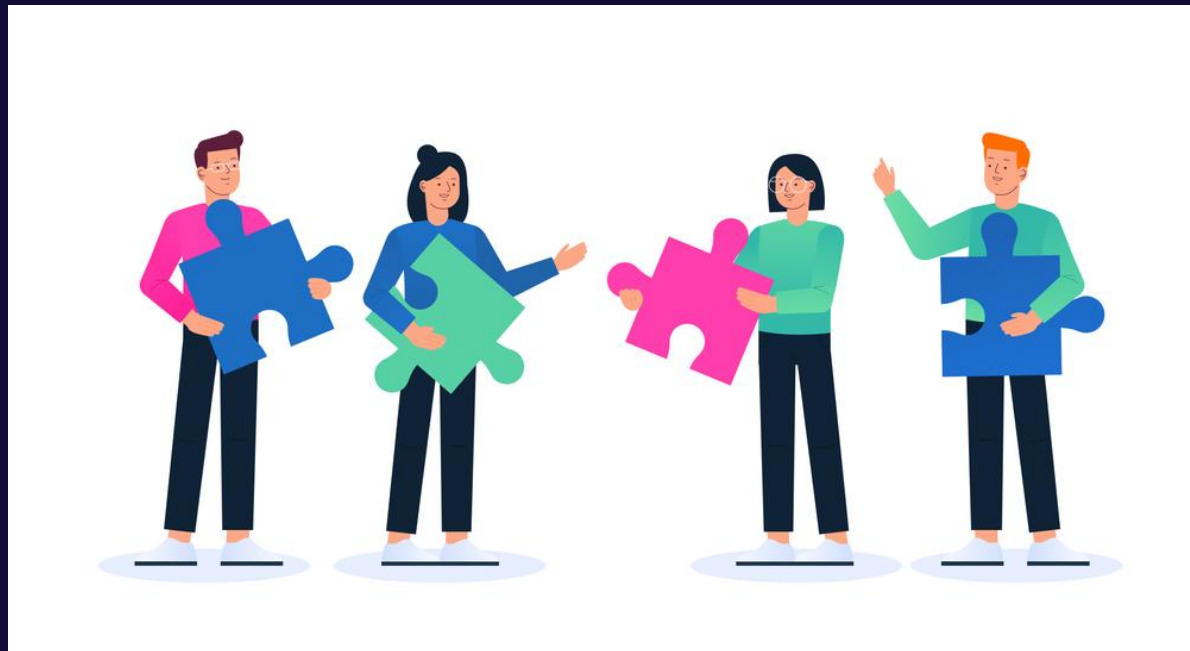
A Quick Exercise

Duration: 3-5 minutes

Task:

- Think about your life journey so far, what you have achieved, what you wanted but couldn't
- Score your achievements in a range of 0-10
- Plot years in X axis and your motivations /feelings in Y axis
- Finally, prepare a Histogram or Line Graph to see your evolution as a person and your years of motivations /struggle

Let's Organize Ourselves



- Self-Organizing Team
- We Create Our Ground Rules
- Values: Respect, Openness, Accountability, Collaboration, Learning Together
- Mark Distribution: Attendance, CT, Assignment, Mid Term, Final

Shaping Tomorrow's Leaders: Industry & Operational Management

Welcome, future innovators! This course is designed to equip you with the essential principles and strategic insights needed to excel in the dynamic worlds of industry and operational management. Prepare to transform your technical prowess into leadership excellence.



Course Objectives & Summary



1. Understand the historical evolution and fundamental principles of industry and operational management.
2. Analyze organizational structures and coordination mechanisms to enhance efficiency and effectiveness.
3. Evaluate personnel management strategies, including motivation, leadership, and performance appraisal, to optimize human resources.
4. Apply cost and financial management techniques, such as break-even analysis and budgetary control, to ensure financial sustainability.
5. Explore marketing and technology management concepts to develop strategies for innovation and competitive advantage.

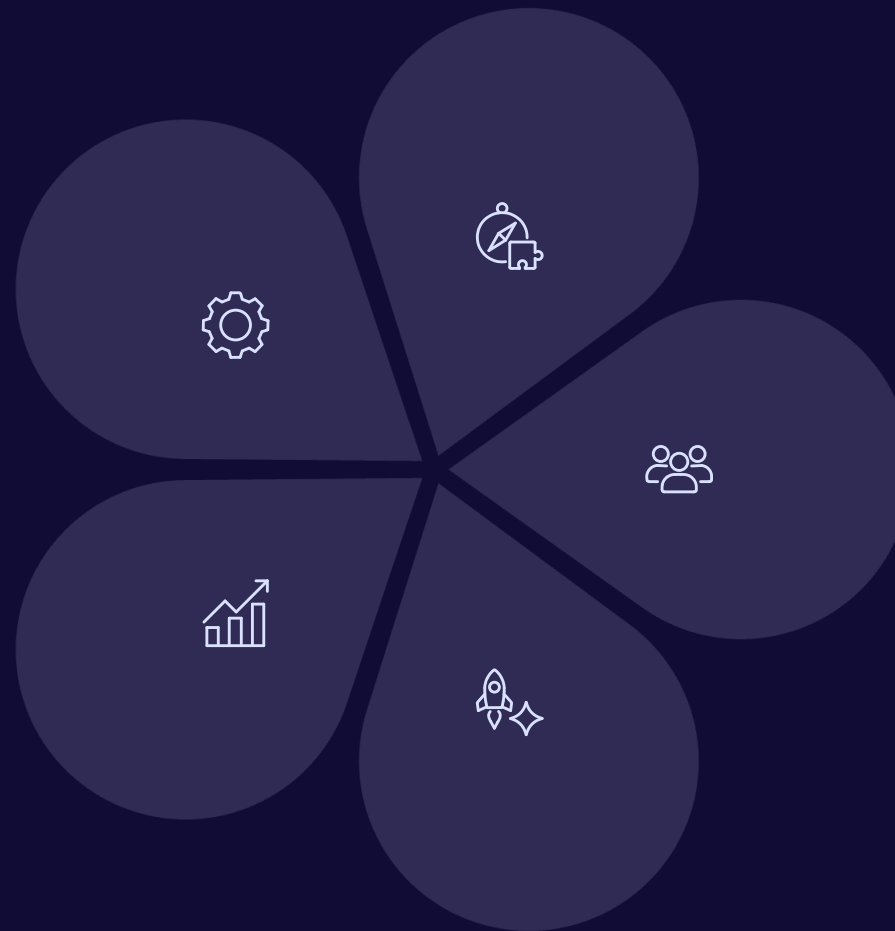
Why Industry & Operational Management?

Operational Efficiency

Optimizing processes for maximum output.

Data-Driven Decisions

Using insights to drive growth.



Strategic Planning

Aligning operations with business goals.

Team Leadership

Fostering high-performing environments.

Digital Transformation

Leveraging tech for competitive advantage.

Textbooks

1. "Operations Management: Sustainability and Supply Chain Management" by Jay Heizer, Barry Render, and Chuck Munson
2. "Project Management: A Systems Approach to Planning, Scheduling, and Controlling" by Harold Kerzner
3. "Human Resource Management" by Gary Dessler
4. "Supply Chain Management: Strategy, Planning, and Operation" by Sunil Chopra and Peter Meindl
5. "Lean Software Development: An Agile Toolkit" by Mary Poppendieck and Tom Poppendieck

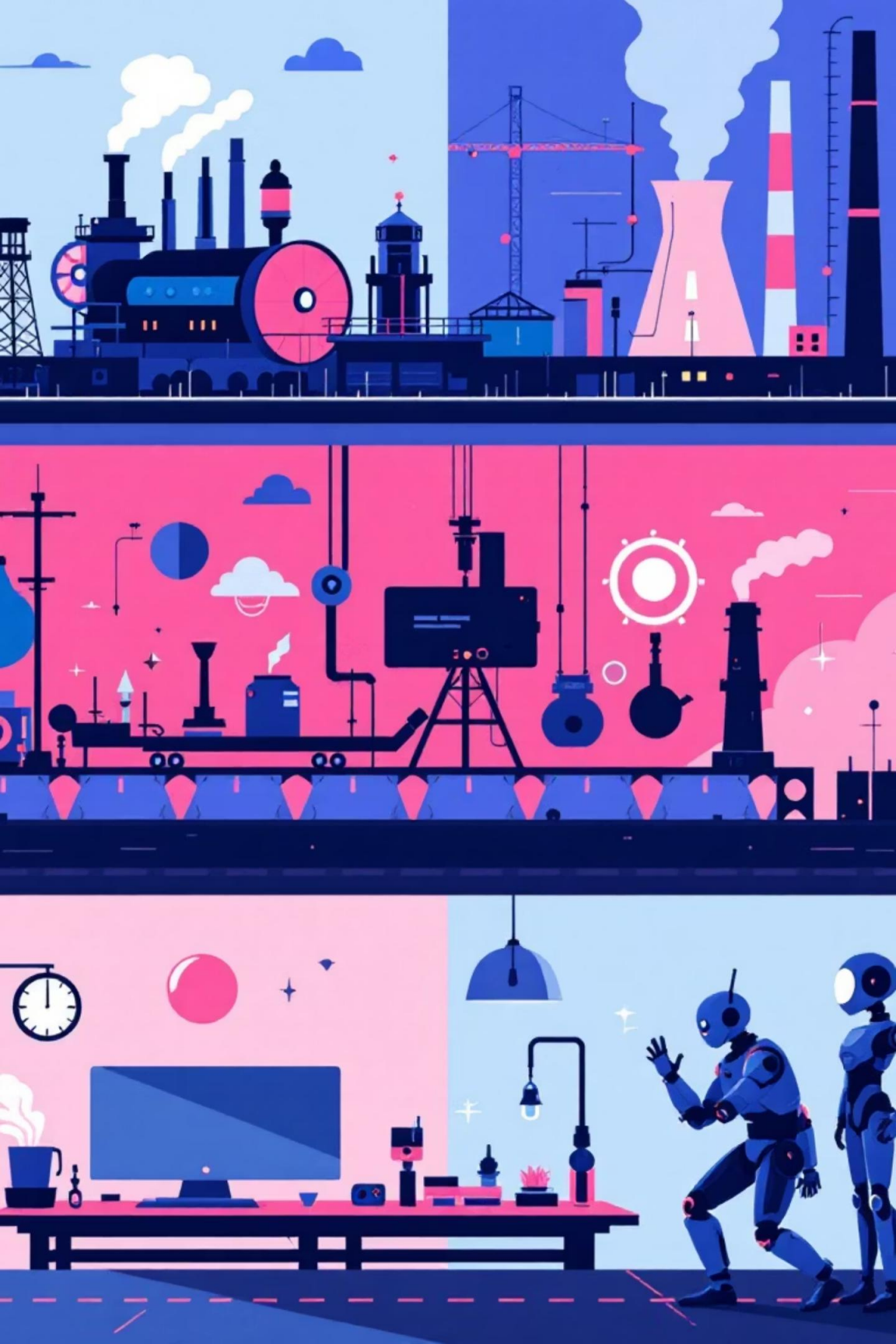
Reference Books & Materials

6. "Operations Management" by William J. Stevenson

Setting the Stage: Pre-Industrial Society (Before 1760)

- Life was predominantly rural, with economies centered around agriculture and manual labor.
- Goods were crafted by hand in homes or small workshops, limiting production scale.
- Energy sources were rudimentary, relying on human, animal, and basic water power.
- Production was slow, seasonal, and highly dependent on natural resources and daylight.
- Britain, with its growing population and burgeoning trade networks, was ripe for transformative change.





The Industrial Revolutions: Transforming the World from Steam to Smart Tech



Origins: Mid-18th Century Britain (c. 1760s–1800)

The Industrial Revolution began in Britain, propelled by significant advancements in agriculture and key technological innovations.

Agricultural Revolution

Improved crop yields led to population growth, providing both a workforce and consumers for new industries.

Spinning Jenny

James Hargreaves' 1764 invention revolutionized textile production, making spinning much faster.

Steam Engine

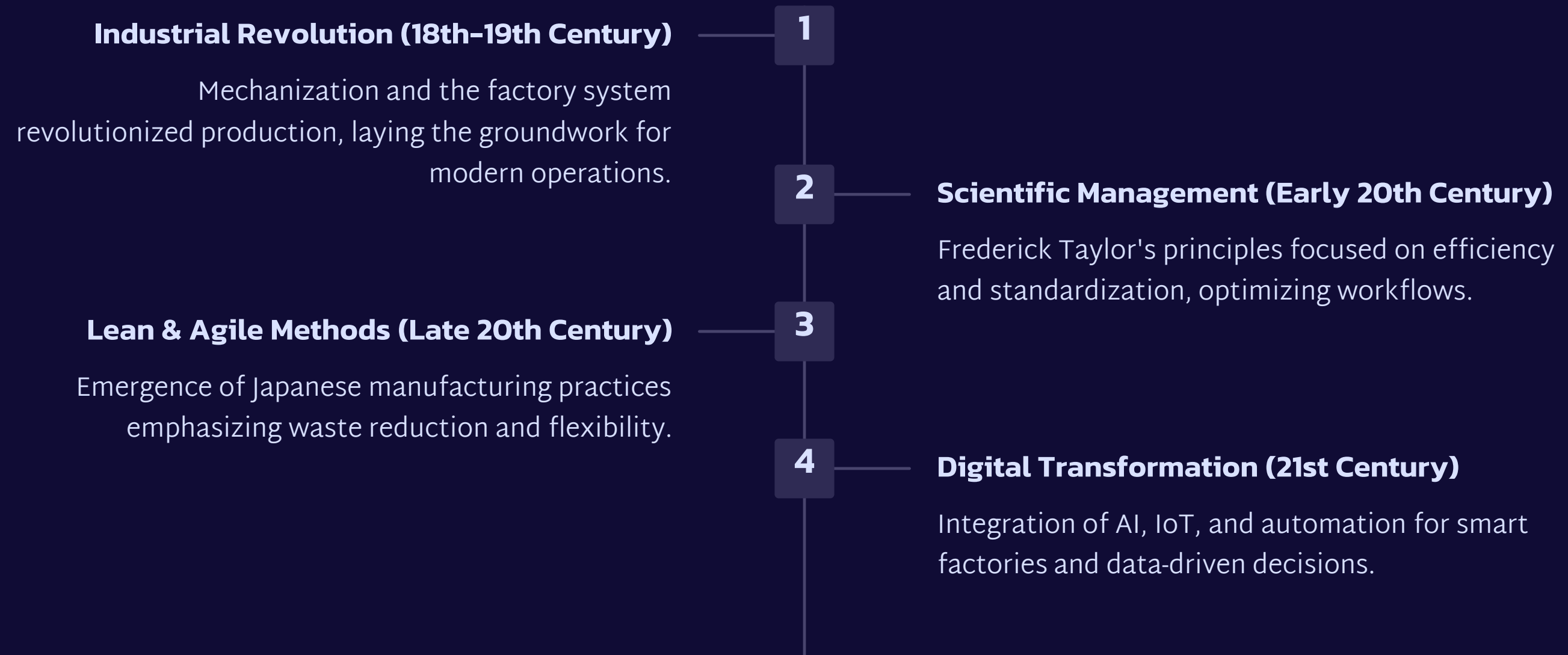
James Watt's 1769 improvements to the steam engine provided a versatile power source for factories and early transportation.

Natural Resources

Britain's abundant coal and iron resources were crucial, fueling the widespread adoption of mechanization.

Tracing the Roots: Evolution of Industry & Operations

Understanding where we come from is crucial to navigating where we're going. The key milestones that have shaped modern industrial practices and operational methodologies.



Industry 1.0: The Mechanical Revolution (c. 1760–1840)



Steam Power & Mechanization

Introduction of steam power revolutionized manufacturing and transport, enabling mass production for the first time.



Key Inventions

Innovations like James Watt's steam engine, the Spinning Jenny, and the power loom spurred unprecedented industrial growth.



Factory System

A significant shift from agrarian economies to mechanized factories, centralizing production and labor.

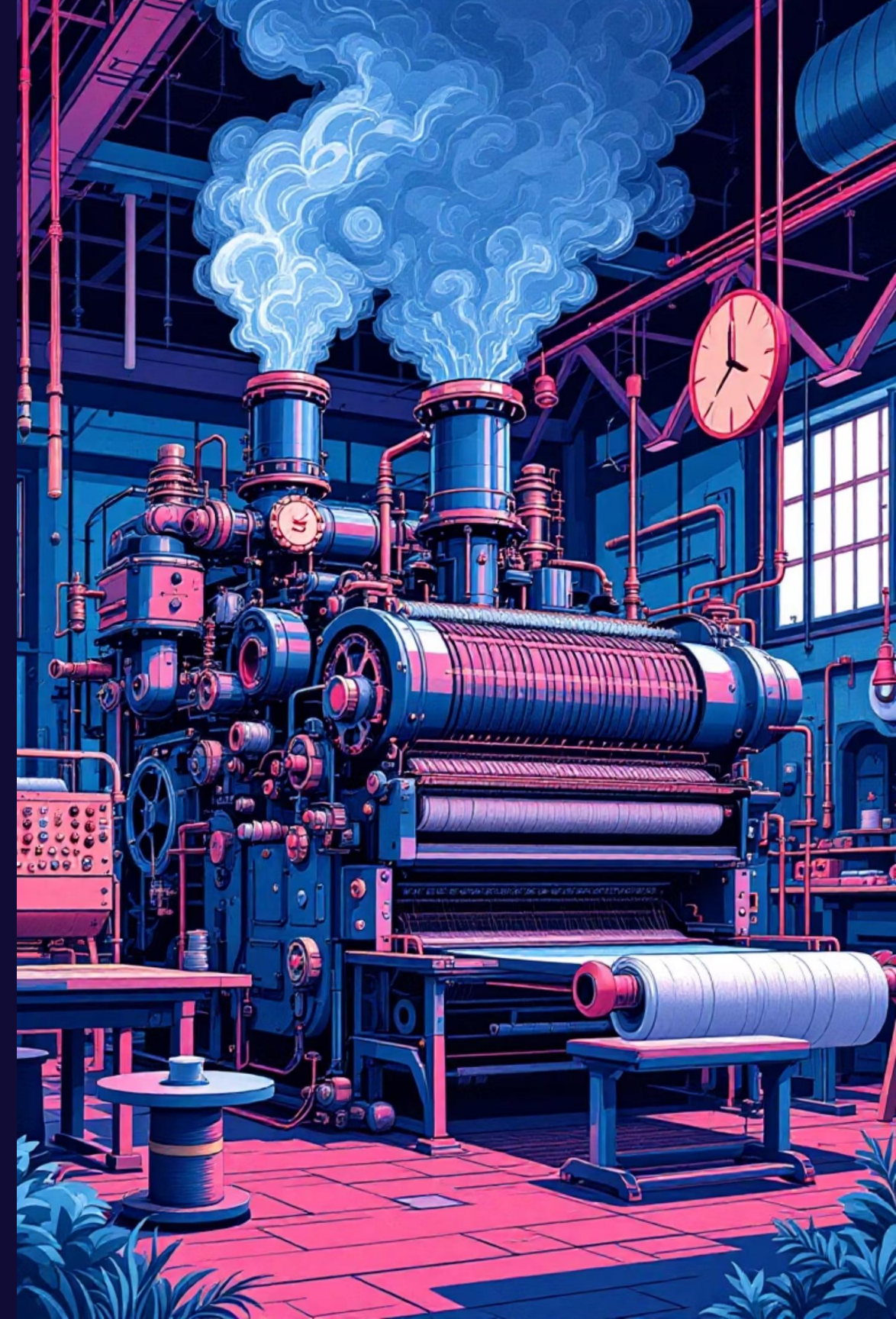


Expanded Markets

Railroads and steamships drastically expanded markets and facilitated rapid urbanization, connecting distant regions.

First Industrial Revolution (1760–1840): The Age of Steam and Textiles

- 1 — 1733: John Kay's Flying Shuttle**
Dramatically sped up weaving, leading to increased demand for yarn.
- 2 — 1764: Spinning Jenny**
James Hargreaves' invention revolutionized yarn production, allowing multiple spools to be spun simultaneously.
- 3 — 1776: James Watt's Steam Engine**
Improved efficiency transformed factories, powering machinery and revolutionizing transportation by land and sea.
- 4 — Urbanization Accelerates**
Factory systems replaced home production, drawing populations to new industrial cities.





Key Technologies That Changed Everything



Steam Engine

Powered trains, steamships, and various factory machinery, making transportation and production more efficient.



Power Loom

Mechanized weaving, significantly boosting textile output and reducing manual labor.



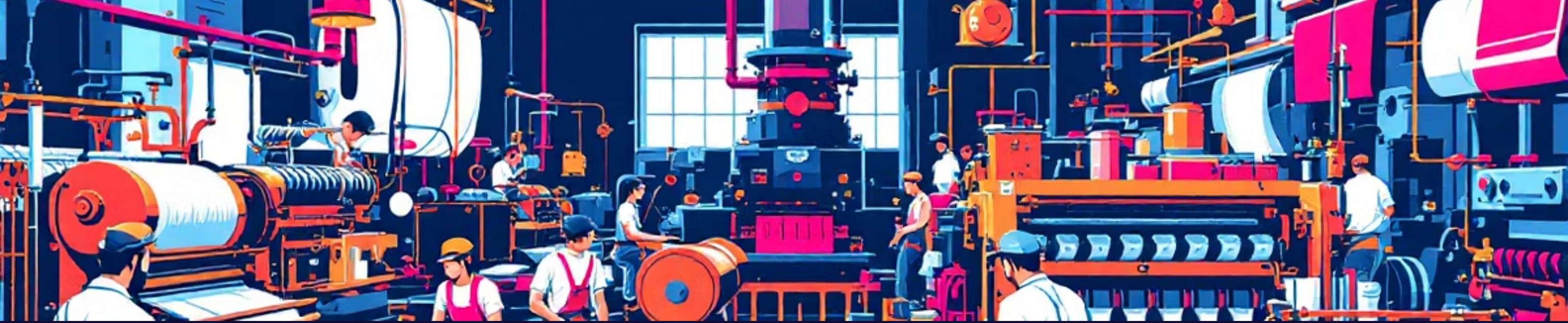
Cotton Gin

(Eli Whitney, 1793) Sped up cotton processing, fueling the textile industry's rapid expansion.



Telegraph

Enabled near-instant communication across vast distances, revolutionizing business and information exchange.



From Handcraft to Machine Power

The First Industrial Revolution marked a profound shift, moving production from artisanal workshops to large-scale, machine-driven factories. Steam power was the driving force, transforming how goods were made and transported.

Industry 2.0: The Mass Production Revolution (c. 1870–1914)

- Electricity and the internal combustion engine enabled faster, large-scale production, dramatically increasing efficiency.
- Groundbreaking innovations included Henry Ford's assembly line, Alexander Graham Bell's telephone, and widespread electric lighting.
- This era saw the rapid rise of foundational industries such as steel and chemicals, shaping modern infrastructure.
- Global connectivity began its expansion with the widespread adoption of the telegraph and the advent of radio technology.



Second Industrial Revolution (1870–1914): Electrification and Mass Production

01

Rise of Electricity

Factories and cities illuminated; Thomas Edison's incandescent lightbulb (1879) transformed daily life.

02

Steel Production Revolution

The Bessemer process enabled cheap, mass production of steel, essential for railroads, bridges, and skyscrapers.

03

Internal Combustion Engine

Led to the development of automobiles and powered flight, drastically changing transportation and travel.

04

Global Communication

The telephone connected voices globally, transforming personal and business communication.



Later 19th Century: Industrial Maturity & Global Spread (1850–1900)

The later half of the 19th century saw industrial processes mature, spread globally, and fundamentally reshape societies.

Bessemer Steel Process (1855)

Mass production of steel made skyscrapers, railways, and bridges feasible.



Alfred Nobel's Dynamite (1866)

Revolutionized mining, quarrying, and construction with powerful explosives.

Global Expansion

Industrialization extended beyond Britain, reaching Europe, North America, and other parts of the world.



Societal Reshaping

The growth of consumer goods, extensive rail networks, and expanding urban centers transformed daily life.

Industrial Revolution's Impact on Society and Economy

- **Economic Transformation:** Shifted from agrarian to industrial economies, leading to the rise of factory labor.
- **Urbanization:** Massive population explosion in cities, giving rise to new middle and working classes.
- **Social Challenges:** Harsh working conditions, child labor, and poor sanitation sparked early labor movements and unions.
- **Global Trade Expansion:** Increased production capabilities led to a significant expansion of global trade, with raw materials sourced worldwide.





Foundations for Computing and Modern Engineering

Precision Manufacturing

The demand for interchangeable parts and mass production techniques laid the groundwork for complex electronics manufacturing.

Communication Networks

Early telegraph and telephone networks inspired and informed the foundational principles of modern communication protocols and data transmission.

Power Systems

Steam and electric power principles were crucial in the design and operation of early computer hardware, providing the necessary energy for computation.

Problem-Solving Mindset

The Industrial Revolution fostered a culture of innovation and systematic problem-solving, which is fundamental to technological advancement in CSE today.

Industry 3.0: The Digital Revolution (c. 1969–2000)



Computing Era

The advent of semiconductors, personal computers, and the internet fundamentally changed information processing and communication.



Advanced Automation

Automation and precision manufacturing techniques transformed production processes, leading to higher efficiency and product quality.



Digital Connectivity

Expansion of global supply chains and digital communication networks fostered unprecedented worldwide collaboration and commerce.



Analog to Digital

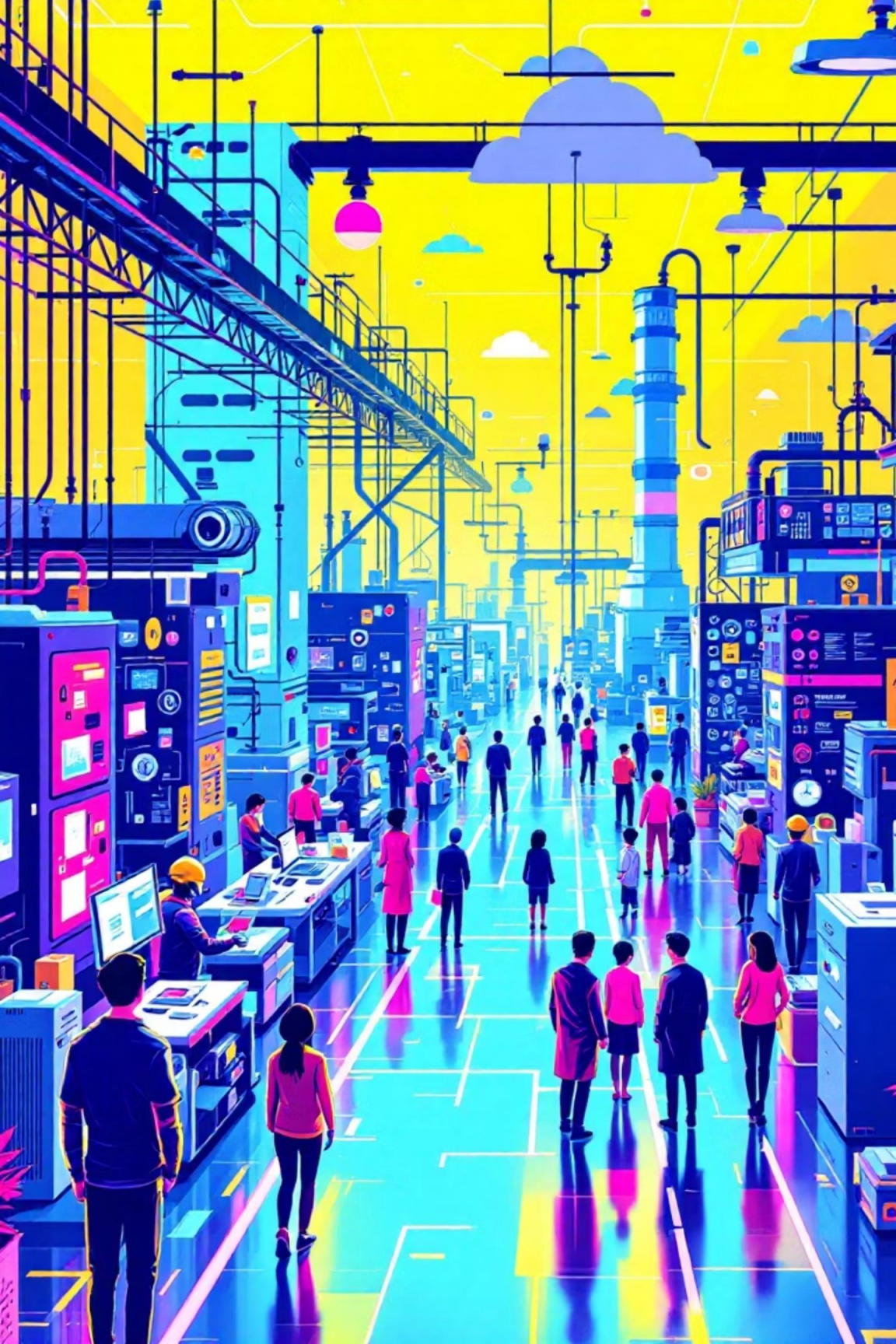
A profound shift from analog to digital technologies reshaped every aspect of business, society, and daily life.

Industry 4.0: The Intelligence Revolution (2000s–Present)

This revolution is characterized by the deep integration of advanced technologies such as Artificial Intelligence (AI), robotics, the Internet of Things (IoT), and big data analytics into industrial processes.

It ushers in "smart factories" where autonomous systems, 3D printing, and interconnected machinery optimize production in real-time.

- Predictive maintenance systems minimize downtime, and personalized manufacturing caters to individual customer needs.
- Key examples include autonomous vehicles, sophisticated robotic automation in logistics, and pervasive virtual assistants in daily life.



Factories Get Smarter: AI and Automation Lead the Way

Industry 4.0 leverages the power of AI, robotics, and connected devices to create highly efficient, adaptive, and intelligent production environments.

Industry 5.0: The Human-Centered Revolution (Emerging)

Human-Machine Collaboration

Focuses on synergistic collaboration between humans and intelligent machines, enhancing human capabilities.

Emerging Trends

Includes human-robot collaboration, eco-friendly manufacturing processes, and the development of ethical AI frameworks.



Sustainability & Ethics

Emphasizes sustainability, inclusivity, and social responsibility in technological development and deployment.

Beyond Efficiency

Technologies are designed to serve not only economic efficiency but also human creativity, well-being, and societal good.



The Impact Across Revolutions: Society & Economy

1st Revolution

Triggered massive urbanization and fundamentally new labor dynamics, setting the stage for modern industrial society.

1

2

2nd Revolution

Mass production fueled the rise of consumer culture and expanded global trade networks on an unprecedented scale.

3

3rd & 4th Revolutions

Digital and intelligent revolutions profoundly reshaped work, raised new questions about privacy, and created hyper-global connectivity.

4

5th Revolution

Aims to balance advanced technology with core human values, environmental care, and a focus on societal benefit.



The Industrial Revolution's Legacy

The profound impact of the Industrial Revolution continues to shape our world today.

Economic Transformation

Shifted economies from agrarian bases to industrial powerhouses, creating new markets and global trade.

Modern Foundations

Laid the groundwork for contemporary manufacturing, advanced transportation systems, and urbanized lifestyles.

Enduring Influence

Sparked social, economic, and political shifts whose effects are still evident in today's global society.

Challenges and Lessons from the Industrial Revolution



- **Environmental Degradation:** Widespread pollution and urban overcrowding were significant consequences.
- **Social Disparity:** Increased social inequalities and severe labor exploitation highlighted the urgent need for systemic reforms.
- **Human-Centric Design:** Emphasized the critical importance of balancing rapid technological progress with human welfare and ethical considerations.
- **Modern Relevance:** These challenges continue to inspire modern sustainability efforts and the development of ethical engineering practices in CSE.

The Industrial Revolution — The Dawn of the Modern Age

This era was a pivotal turning point, fundamentally reshaping human society through relentless innovation and industrial expansion.

- Its groundbreaking inventions and transformative ideas continue to influence global technology, economies, and cultures.
- Understanding this rich timeline provides invaluable insight into the origins of our industrialized world and its ongoing evolution.





Looking Ahead: Preparing for the Future of Industry

The journey through the Industrial Revolutions reveals a continuous cycle of transformation, driven by relentless innovation and human ingenuity.

→ **Continuous Transformation**

History shows that industry is always evolving, pushing boundaries of what's possible.

→ **Rethinking Technology's Role**

Industry 5.0 challenges us to define technology's purpose beyond mere efficiency.

→ **Human-Machine Synergy**

Embracing collaboration unlocks sustainable growth and addresses complex global challenges.

→ **Integrating with Humanity**

The future depends on our ability to merge technological advancement with human values.



Conclusion: The Industrial Revolution's Legacy for CSE Students

→ **Foundation of Modernity**

It fundamentally transformed how we build, communicate, and innovate, laying the very roots of modern computing and engineering.

→ **Appreciating Evolution**

Understanding this pivotal history helps us truly appreciate the complex evolution and trajectory of technology to where it is today.

→ **Giants' Shoulders**

Today's digital revolution stands firmly on the shoulders of these industrial giants, leveraging their innovations and insights.

→ **Your Future Role**

As CSE students, your role is to drive the next wave of innovation, responsibly and creatively, learning from the past to build a better future.

Thank you